Digit Test Process

The CRT monitors were benchmarked on different parameters based on the tests carried out by DisplayMate, a dedicated video utility for setting up, tuning-up, calibrating, evaluating and testing video displays, monitors and even complete video systems.

It is the only utility devoted to monitors and video boards. The tests carried out by DisplayMate were:

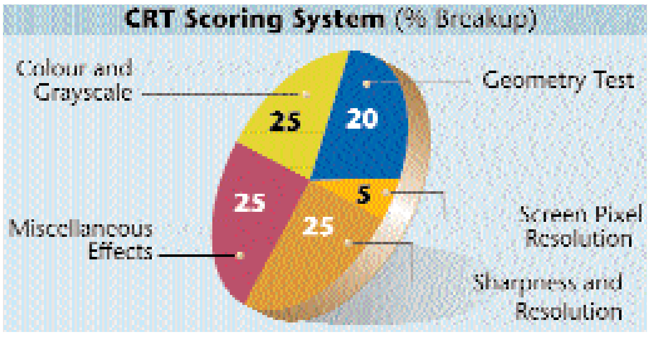
1. Geometry test: It comprises six subtests. The pincushion and barrel distortion test shows the monitor's capacity to display straight lines; the screen regulation tests examine the extent to which an image expands in bright areas on the screen, and contracts in the areas which are dim.

2. Sharpness and Resolution test: This includes seven subtests that examine the sharpness of horizontal as well as vertical resolutions of the images, variations from optimum focus over the screen, sharpness and resolution of the entire screen, the visible RGB component and moiré pattern.

3. Screen pixel resolution: A test named Scaled Diamond is used to check for any jaggedness on the screen while displaying straight as well as diagonal lines.

4. Colour and gray scale: This test comprises nine sub-tests that check for any kind of ghosting or streaking.

5. Miscellaneous effects: This is a set of five subtests that check the monitor for flickering and how uniformly the screen is visible while viewing full-screen images.



Before testing, the monitors were configured with the correct set of drivers provided by the vendor or the ones supported by default in Windows. Three reviewers tested all the monitors and three separate test scores were logged for each monitor. The scores were then averaged and compiled to generate an overall score. We switched on the monitors for approximately 10 minutes for warming them up so that each reviewer would get exactly the same results.

To keep the ambient light condition constant for each monitor, they were tested in a closed room with only one incandescent lamp (40 W) on at the back of the monitor. This was also done to reduce any glare on the monitor screen. The door to the room was kept closed and the see- through glass on the door was covered with a black sheet of paper on both sides to keep the room dark. The monitors were tested at the following constant resolutions:

15-inch/17-inch: 1024x768x75Hz

19-inch/21-inch/22-inch: 1280x1080x75Hz

The test bed comprised a P-III 700 MHz processor, 128 MB of RAM. The graphics card was GeForce2 Ultra 64 MB DDR.

Finally, the monitors were checked for other parameters like construction and build quality, ease of use of the OSD, user manual, integration of extras such as USB functionality, speakers, and so on.

considered acceptable. As you can see from the table in one of the following pages, Microtek's entry logged a very poor score of 4.6. On the other hand, the Compaq MV540 scored an amazing 16.6, mowing over the rest of the competition by a fair margin. It also carried with it a pretty hefty price tag. The closest that any monitor could come to it was the Benq V551-M, which would be a decent bargain considering its price. Another rather interesting fact was that only four of the 11 monitors in this category—namely BPL, Viewsonic, Proview and Samsung—provided OSDs; the rest came with analog controls.

Among the 17-inch monitors, the Viewsonic G73f came out on top in the Geometry tests. It had an impressive overall screen geometry score of 20.8 which was over 3 points ahead of its nearest rival, the Samsung SyncMaster 753s and way ahead of the average score for the 17-inch category (13 points). A large component of its good showing in the Geometry test was the impressive scores in the screen regulation test—we observed practically no changes in the size and shape of the image and no perceptible clipping of the screen when brightness levels were suddenly increased. Surprisingly however, the monitor gave only average results when generating parallel line patterns in the distortion test.

Another impressive performer was the Samsung SyncMaster 753s, which showed very little image distortion and came second only because it showed some variation in dimensions during the Screen Regulation test.

The LG Flatron 700s was a big disappointment and gave us way below average results in all four image regulation tests—the variation in the size and shape of the test images was drastic.

In the 19-inch category, Sony CPD-420 showed pretty impressive results. It logged an extremely high score of 24 points—the highest score in the entire comparison test, across all categories of CRT monitors. Close on its heels were the Compaq MV-940 and the Viewsonic E90 with scores of 23.6 and 23.3 points, respectively. The Sony CPD-420 was also the only monitor with an Aperture Grill tube, which uses the patented Trinitron technology. This test clearly indicates that the Sony CPD-420 is ideal for graphic and Web designers who have to deal with a lot of vector images and animations.

The 21- and 22-inch monitors are probably the most widely used for video editing, CAD/CAM applications, graphic designing and engineering drawings. Considering that these monitors incorporate high precision electronics, a score of over 20 in the screen geometry test should be considered a bare minimum.

There was stiff competition in the Geometry test in this category. The Viewsonic G810 showed outstanding results in the Pincushion/Barrel test. It logged an overall score of 23.5 in the Screen Geometry test, beating the supremely elegant Sony CPD-520, which scored 22.8 points.

In contrast, the Benq P211 logged a disappointing score of 14.4 points. The low overall score in this test can be attributed ➜